

Dual-Stream Spatio-Temporal DQN with Temporal Attention for Adaptive Traffic Signal Control

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Abstract—Traffic congestion remains a major challenge, as conventional and adaptive signal control systems often fail to capture the spatio-temporal dependencies of traffic flow. This limitation results in inefficient green phases, increased delays, fuel waste, and excessive emissions. To address this issue, we propose ST-DQN-TA, a dual-stream spatio-temporal Deep Q-Network with attention. The spatial stream models lane-level and global intersection features to form spatial states, while the temporal stream captures historical spatial traffic states using Multi-Head Self-Attention (MHSA). The outputs of these streams are fused through a multi-layer perceptron (MLP) and passed to a dueling DQN head to generate context-aware phase-duration decisions for improved traffic efficiency. Experiments conducted in the SUMO (Simulation of Urban MObility) environment, integrated with the Traffic Control Interface (TraCI) API, on a standard four-way intersection demonstrate that ST-DQN-TA consistently outperforms conventional fixed-time baselines, underscoring the importance of explicit spatio-temporal modeling with temporal attention for adaptive traffic signal control.

Index Terms—Adaptive Traffic Control, Spatio-Temporal Attention, Multi-Head Self-Attention, Dual-Stream Dueling DQN, Traffic flow optimization.

I. INTRODUCTION

Many road networks experience traffic congestion due to static traffic signal timers that do not account for real-time traffic density. This leads to issues such as harsh driving conditions, traffic rule violations, accidents, and increased air pollution [1]. The inefficiency arises because existing traffic light systems often fail to adapt dynamically to the changing traffic conditions often lead to suboptimal phase allocations and unnecessary delays. For instance, they may continue to give a green light to an empty lane while other lanes with heavy traffic remain stopped until the light changes [2]. Such scenarios result in increased travel time, higher vehicle fuel consumption, and greater environmental pollution [3] [4]. Therefore, implementing a system that can adaptively control traffic signal timers is essential for improving traffic flow and

reducing congestion, highlighting the need for an Adaptive Traffic Control System as a practical solution [1].

Adaptive traffic signal control using Deep Reinforcement Learning (DRL) has demonstrated considerable promise by learning real-time control policies directly from vehicular data streams. However, most DRL-based methods rely on either aggregated traffic features or single-stream architectures, which limit their ability to capture complex spatio-temporal dependencies across multiple directions. This can reduce the effectiveness of traffic anticipation and adaptive signal optimization, particularly under highly dynamic traffic conditions.

To address these challenges, we propose the Dual-Stream Spatio-Temporal DQN with Temporal Attention (ST-DQN-TA). The spatial stream models lane-level and global intersection features to form spatial states, while the temporal stream captures historical spatial traffic states using Multi-Head Self-Attention (MHSA). The outputs of these streams are fused through a multi-layer perceptron (MLP) and passed to a dueling DQN head to generate context-aware phase-duration decisions for improved traffic efficiency. The framework is trained in the SUMO [5] simulator via the TraCI [6] interface, enabling real-time state extraction, action execution, and reward-driven learning with experience replay. This framework results in more efficient and adaptive traffic signal control.

II. RELATED WORK

Over the past years, traffic signal control has evolved significantly with the integration of machine learning, particularly reinforcement learning (RL) and attention-based models [8]. These techniques improve adaptability by leveraging real-time data, but important limitations remain.

In [9], the authors introduced Max Queue-Length (M-QL) and AttentionLight. M-QL improves phase selection by prioritizing maximum queues, while AttentionLight employs self-attention within RL to capture correlations between signal

phases. Although these methods improve accuracy, they lack explicit spatio-temporal modeling, resulting in reactive policies that cannot anticipate how traffic conditions evolve over time.

Another effort, AMUW-DDQN [10], incorporates attention and Squeeze-and-Excitation Networks (SENet) to highlight critical features, leading to more accurate value estimation and reduced delay. However, its focus on spatial feature weighting without robust temporal modeling limits adaptability under dynamic traffic demand, leaving long-range dependencies underutilized.

Federated learning has also been applied to improve scalability. For example, Li et al. [11] proposed a federated RL framework based on PPO that enables distributed training across intersections while preserving data privacy. This improves convergence speed and generalization, but the communication and synchronization overhead reduces responsiveness, making it less suitable for single-intersection, real-time control.

Similarly, Yang et al. [12] combined Graph Attention Networks (GATs) with multi-agent RL to capture spatial-temporal relationships across networks. While effective for coordination across multiple intersections, this approach depends heavily on graph structures and introduces significant computational cost, limiting its practicality for latency-sensitive, single-intersection applications.

In contrast, our proposed ST-DQN-TA combines the strengths of these directions while addressing their limitations. By explicitly modeling both spatial states and long-range temporal dependencies through a dual-stream architecture with multi-head self-attention, it balances effectiveness and computational efficiency. Unlike distributed or graph-based approaches, it directly targets spatio-temporal reasoning at the intersection level, enabling robust real-time decision-making without unnecessary overhead.

III. PROPOSED METHODOLOGY

A. Problem Formulation

The traffic setting is a four-way intersection where, unlike fixed-time systems, the next phase is adaptively chosen based on real-time traffic conditions.

The traffic signal control problem is formulated as a Markov Decision Process (MDP) with the following components:

- **State Space ($\mathcal{S}$):** Dual representation consisting of spatial features $\mathbf{s}_t^{\text{spa}} \in \mathbb{R}^{21}$ and temporal sequences $\mathbf{s}_t^{\text{temp}} \in \mathbb{R}^{T \times 21}$ (see Section III-B1 for details).
- **Action Space:** The agent makes decisions by selecting a combination of signal phase and green-time duration. Four signal phases are considered ($N_{\text{phases}} = 4$), and thirty-five possible green durations ($N_{\text{durations}} = 35$) are obtained by dividing the range of 10 s to 180 s into 5-second intervals. This range represents a practical search space for urban signal control, covering both short and long cycles used under varying traffic conditions. The

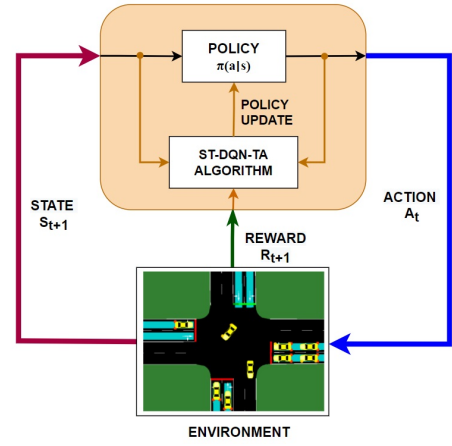


Fig. 1. Reinforcement Learning Framework with the ST-DQN-TA Algorithm for Adaptive Traffic Signal Control.

lower limit of 10 s avoids phases that are too short for safe vehicle clearance, while the upper limit of 180 s accommodates high-demand conditions. The 5-second step size was selected to balance control resolution with computational efficiency. A smaller step (e.g., 1 s) would substantially enlarge the action space and increase training time, whereas the 5 s increment provides sufficient timing precision while keeping the search space tractable. A similar discretization approach was adopted in recent work on enhanced deep reinforcement learning for adaptive traffic signal control [13], where green-time actions were also defined in 5-second increments. Hence, the total number of discrete actions is:

$$|\mathcal{A}| = N_{\text{phases}} \times N_{\text{durations}} = 4 \times 35 = 140. \quad (1)$$

B. Dual-Stream Spatio-Temporal DQN with Attention

1) *State Representation:* The state space consists of spatial and temporal components capturing traffic dynamics across four intersection approaches.

Spatial State: The spatial state vector at time t captures current traffic conditions and contains direction-specific and global features:

$$\mathbf{s}_t^{\text{spa}} = \begin{bmatrix} \mathbf{d}_{\text{west}}; \mathbf{d}_{\text{east}}; \mathbf{d}_{\text{north}}; \mathbf{d}_{\text{south}}; \\ Q_{\text{total}}; N_{\text{total}}; \phi_{\text{current}}; t_{\text{phase}}; \phi_{\text{last}} \end{bmatrix} \in \mathbb{R}^{21}, \quad (2)$$

where each direction vector $\mathbf{d}_i \in \mathbb{R}^4$ represents per-approach traffic metrics.

For each direction $i \in \{\text{west, east, north, south}\}$:

$$\mathbf{d}_i = [Q_i; O_i; v_i; n_i] \in \mathbb{R}^4, \quad (3)$$

where:

- Q_i : Queue length in direction i
- O_i : Lane occupancy in direction i
- v_i : Mean speed in direction i
- n_i : Vehicle count in direction i

Temporal State: The temporal state maintains a sliding window history of previous T spatial states:

$$\mathbf{s}_t^{\text{temp}} = [\mathbf{s}_{t-T+1}^{\text{spa}}; \mathbf{s}_{t-T+2}^{\text{spa}}; \dots; \mathbf{s}_t^{\text{spa}}] \in \mathbb{R}^{T \times 21}. \quad (4)$$

2) *Dual-Stream Spatio-temporal Processing:* Our proposed adaptive traffic control architecture processes spatial and temporal traffic information through separate parallel streams before fusion, enabling the system to analyze both current traffic conditions and historical evolution patterns. We use a standard 8-head attention mechanism [7], with head dimensions $d_k = d_v = 8$, within the temporal stream. This allows each head to focus on different aspects of traffic dynamics, from short-term fluctuations to long-range flow correlations.

Spatial Stream: For a batch of spatial states $\mathbf{S}_t^{\text{spa}} \in \mathbb{R}^{B \times 21}$, the input features are projected into an embedding space of dimension $d_{\text{model}} = 64$. The embedding is computed with dropout ($p = 0.2$) as:

$$\mathbf{H}_{\text{spa}} = \text{Dropout}_{0.2}(\text{ReLU}(\mathbf{S}_t^{\text{spa}} \mathbf{W}^{\text{spa}} + \mathbf{b}^{\text{spa}})) \in \mathbb{R}^{B \times 64}, \quad (5)$$

where $\mathbf{W}^{\text{spa}} \in \mathbb{R}^{21 \times 64}$ and $\mathbf{b}^{\text{spa}} \in \mathbb{R}^{64}$.

Temporal Stream: The temporal input is represented as:

$$\mathbf{S}_t^{\text{temp}} \in \mathbb{R}^{B \times T \times 21},$$

where B is the batch size, T is the sequence length, and 21 is the number of features per time step.

To project the raw features into a higher-dimensional latent space, a linear embedding is applied:

$$\mathbf{H}_{\text{embed}}^{\text{temp}} = \mathbf{S}_t^{\text{temp}} \mathbf{W}^{\text{temp}} + \mathbf{b}^{\text{temp}} \in \mathbb{R}^{B \times T \times 64}, \quad (6)$$

with learnable parameters $\mathbf{W}^{\text{temp}} \in \mathbb{R}^{21 \times 64}$ and $\mathbf{b}^{\text{temp}} \in \mathbb{R}^{64}$.

To encode temporal order, positional encoding $\mathbf{PE} \in \mathbb{R}^{T \times 64}$ is added:

$$\mathbf{H}_{\text{embed}}^{\text{temp,PE}} = \mathbf{H}_{\text{embed}}^{\text{temp}} + \mathbf{PE} \in \mathbb{R}^{B \times T \times 64}. \quad (7)$$

Finally, a multi-head self-attention (MHSA) module captures temporal dependencies:

$$\mathbf{H}_{\text{MHSA}}^{\text{temp}} = \text{MHSA}(\mathbf{H}_{\text{embed}}^{\text{temp,PE}}) \in \mathbb{R}^{B \times T \times 64}. \quad (8)$$

After pooling and normalization:

$$\mathbf{h}_{\text{temp}} = \text{LNorm} \left(\frac{1}{T} \sum_{\tau=1}^T \mathbf{H}_{\text{MHSA}}^{\text{temp}}[:, \tau, :] \right) \in \mathbb{R}^{B \times 64}. \quad (9)$$

3) *Feature Fusion and Processing:* The fused features are obtained through concatenation:

$$\mathbf{F}_{\text{fused}} = [\mathbf{H}_{\text{spa}}; \mathbf{h}_{\text{temp}}] \in \mathbb{R}^{B \times 128}. \quad (10)$$

A two-layer MLP with dropout ($p = 0.2$) processes $\mathbf{F}_{\text{fused}}$ to learn higher-level feature representations. We define the hidden layer dimensions as $H_1 = 256$ and $H_2 = 128$:

$$\mathbf{H}_1 = \text{Dropout}_{0.2}(\text{ReLU}(\mathbf{F}_{\text{fused}} \mathbf{W}_1 + \mathbf{b}_1)) \in \mathbb{R}^{B \times 256}, \quad (11)$$

where $\mathbf{W}_1 \in \mathbb{R}^{128 \times 256}$ and $\mathbf{b}_1 \in \mathbb{R}^{256}$.

$$\mathbf{H}_{\text{fused}} = \text{Dropout}_{0.2}(\text{ReLU}(\mathbf{H}_1 \mathbf{W}_2 + \mathbf{b}_2)) \in \mathbb{R}^{B \times 128}, \quad (12)$$

where $\mathbf{W}_2 \in \mathbb{R}^{256 \times 128}$ and $\mathbf{b}_2 \in \mathbb{R}^{128}$.

The first layer expands the feature space from 128 to 256 dimensions to capture more complex patterns, while the second layer compresses back to 128 dimensions for computational efficiency. Thus, $\mathbf{H}_{\text{fused}} \in \mathbb{R}^{B \times 128}$ serves as the final latent representation for the dueling DQN heads.

4) *Dueling DQN Output Architecture:* We adopt a dueling DQN structure that separates value and advantage functions:

Value Function:

$$V(\mathbf{s}_t) = \mathbf{H}_{\text{fused}} \mathbf{w}_V + b_V \in \mathbb{R}^{B \times 1}, \quad (13)$$

where $\mathbf{w}_V \in \mathbb{R}^{128 \times 1}$ and $b_V \in \mathbb{R}$.

Advantage Function:

$$A(\mathbf{s}_t, \mathbf{a}) = \mathbf{H}_{\text{fused}} \mathbf{W}_A + \mathbf{b}_A \in \mathbb{R}^{B \times |\mathcal{A}|}, \quad (14)$$

where $\mathbf{W}_A \in \mathbb{R}^{128 \times |\mathcal{A}|}$ and $\mathbf{b}_A \in \mathbb{R}^{|\mathcal{A}|}$.

Q-Function:

$$Q(\mathbf{s}_t, \mathbf{a}) = V(\mathbf{s}_t) + A(\mathbf{s}_t, \mathbf{a}) - \frac{1}{|\mathcal{A}|} \sum_{a' \in \mathcal{A}} A(\mathbf{s}_t, a'), \quad Q \in \mathbb{R}^{B \times |\mathcal{A}|}. \quad (15)$$

where $V(\mathbf{s}_t)$ is broadcast across the action dimension for element-wise addition.

C. Training Framework with SUMO

The ST-DQN-TA agent is trained in SUMO via TraCI in a closed-loop setup. At each step, the agent observes the state s_t , which includes lane-level and aggregated intersection features along with a temporal sequence of length $T = 10$ to capture recent traffic dynamics. This value balances temporal context with computational cost. Based on this input, the agent selects an action a_t using an ϵ -greedy policy over the dueling Q-head, applies it to the signal controller, advances the simulation, and receives the reward r_t and next state s_{t+1} . Transitions (s_t, a_t, r_t, s_{t+1}) are stored in a prioritized replay buffer.

1) *Reward Function:* The reward function r_t is designed to balance multiple objectives at the intersection: maximizing traffic flow, minimizing vehicle queues and waiting time, reducing fuel consumption, and maintaining balanced queues across directions. In this work, we define the reward at time t as:

$$r_t = w_{\text{flow}} \sum_d \bar{v}_d - w_{\text{queue}} Q_{\text{total}} - w_{\text{wait}} H_{\text{current}} - w_{\text{fuel}} \sum_d F_d - w_{\text{balance}} \text{std}(Q_d^{\text{final}}) \quad (16)$$

where $\bar{v}_d$ is the mean vehicle speed at direction d , Q_{total} is the total queue length, H_{current} is the number of halted vehicles (speed < 0.1 m/s), F_d is fuel consumption at detector d in milliliters, and $\text{std}(Q_d^{\text{final}})$ is the standard deviation of queue lengths across directions.

Table I shows the weight coefficients and their corresponding components in the reward function. These values were

TABLE I
REWARD FUNCTION WEIGHT COEFFICIENTS

Component	Weight	Description
w_{flow}	0.125	Flow reward per m/s per direction
w_{queue}	0.02	Queue penalty per vehicle
w_{wait}	0.05	Penalty per halted vehicle
w_{fuel}	0.001	Fuel penalty per milliliter consumed
w_{balance}	0.5	Queue imbalance penalty (std. deviation)

tuned empirically to balance the trade-offs between flow optimization, queue management, fuel efficiency, and directional balance. The agent successfully learns to minimize queuing and waiting time by exploiting the natural correlation between traffic flow and queue reduction, as demonstrated in the results (Section IV).

Learning proceeds via minibatch updates with a periodically synchronized target network. The temporal-difference (TD) target y_t and the loss function $\mathcal{L}(\theta)$ are defined as:

$$y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-), \quad (17)$$

$$\mathcal{L}(\theta) = \mathbb{E} \left[w_t (y_t - Q(s_t, a_t; \theta))^2 \right], \quad (18)$$

where θ and θ^- denote the online and target network parameters, and w_t is the importance-sampling weight that corrects for prioritized sampling.

2) *Computational Setup and Hyperparameters*: All experiments were conducted on a laptop powered by an Intel Core i7 processor with 32 GB of RAM and equipped with an NVIDIA GeForce GTX 1660 Ti GPU (6 GB GDDR6 VRAM). The agent was trained online in the SUMO traffic simulator via the TraCI interface. A complete training run comprising 250 episodes required approximately 19.9 hours to converge, with checkpoints saved every 50 episodes and performance metrics recorded only at these intervals to minimize computational overhead.

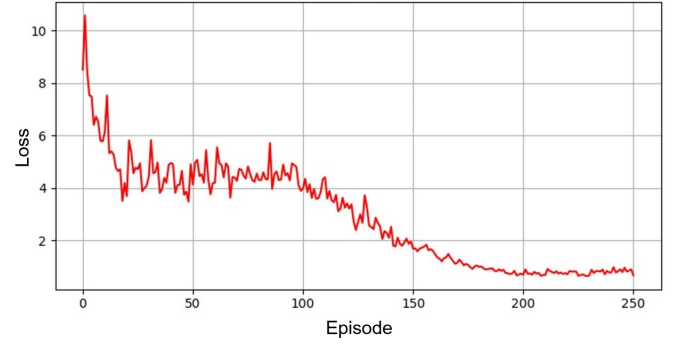
To balance training efficiency and learning stability under these computational constraints, several key hyperparameters were carefully tuned. Table II summarizes the main hyperparameter settings used during training.

IV. RESULTS AND DISCUSSION

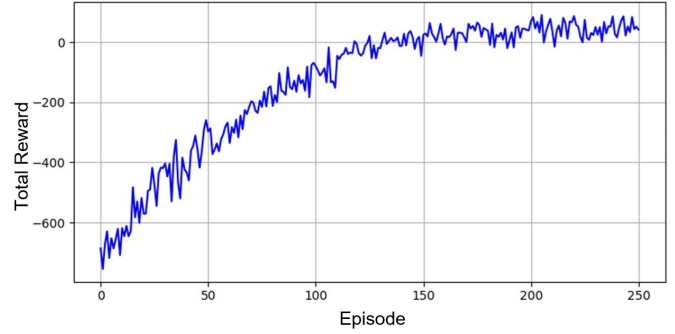
This section evaluates the ST-DQN-TA architecture for adaptive traffic signal control at a four-way intersection, focusing on *training dynamics*, *convergence*, and *traffic performance*.

TABLE II
HYPERPARAMETERS USED FOR TRAINING THE ST-DQN-TA AGENT

Parameter	Value
Number of episode	250
Max Steps per Episode	100
Batch size	32
Prioritized Experience Replay size	10000
Target Network Update	Every 10 episodes
Replay frequency	Every 4 steps
Learning rate (α)	0.0005
Discount factor (γ)	0.99
Exploration (ϵ -greedy)	1.0 $\rightarrow$ 0.05 (decay 0.02)



(a) Training loss over episodes.



(b) Total reward progression.

Fig. 2. Convergence behavior of ST-DQN-TA: complementary trends in loss and reward.

A. Training Dynamics and Convergence

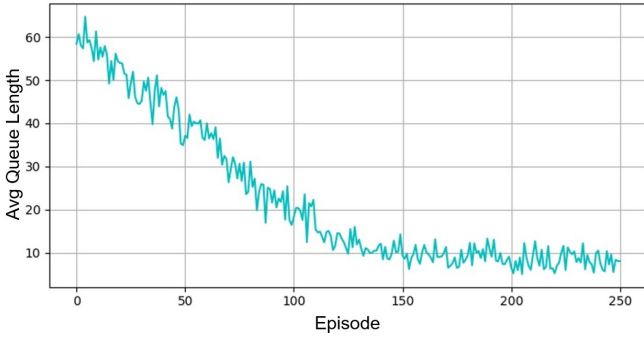
1) *Loss evolution*: As shown in Fig. 2a, the loss starts near 10, drops sharply in the first 20 episodes, oscillates between 3–6, and then steadily declines to around 1.0 by episode 200, indicating stable policy updates.

2) *Reward progression*: In Fig. 2b, rewards begin near -700 , rise steadily, cross into positive territory around episode 150, and stabilize between 0 and 100. Together with the loss trend (Fig. 2a), this confirms successful convergence.

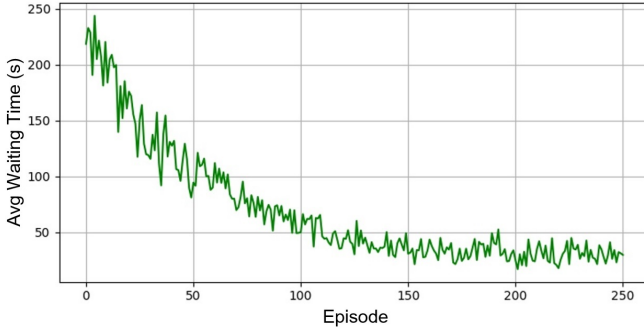
B. Traffic Performance Metrics

1) *Queue Length and Delay*: As shown in Fig. 3, both metrics decrease sharply during early training. Average waiting time falls from over 220 s to below 50 s by episode 120, stabilizing around 30–40 s thereafter. Similarly, average queue length drops from above 60 vehicles to below 20 by episode 100, converging to 7–12 vehicles after episode 150. These trends demonstrate that ST-DQN-TA effectively reduces congestion and delays as training progresses.

2) *Directional Fairness*: As shown in Fig. 4, all four approaches initially experience queues of 10–20 vehicles, with west occasionally peaking above 20. By episode 100, queues across directions fall below 10, and after 150 episodes, they converge to 2–4 vehicles. The close overlap of the curves indicates that ST-DQN-TA achieves balanced phase allocation and equitable performance across directions.



(a) Average queue length (over 250 episodes).



(b) Average waiting time (over 250 episodes).

Fig. 3. Queue and delay trends learned by ST-DQN-TA.

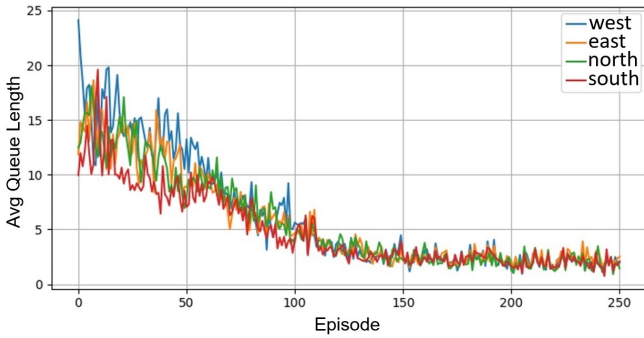


Fig. 4. Directional average queue length across directions(over 250 episodes).

3) *Fuel Consumption*: Average fuel consumption per vehicle is initially high and volatile, then declines and stabilizes around 0.02–0.04L after ~ 150 episodes (Fig. 5). This indicates fewer unnecessary stops and smoother flows enabled by the learned control.

4) *Performance Comparison*: To evaluate the effectiveness of the RL-based ST-DQN-TA controller, we compared its performance against fixed-time baselines with cycle lengths of 30–60 seconds. The results are shown in Table III.

As summarized in Table III, the ST-DQN-TA model significantly outperforms all fixed-time baselines. Compared to the best-performing baseline (30s cycle), it achieves a $\sim 79\%$ reduction in average queue length, a $\sim 66\%$ reduction in average waiting time, and a $\sim 63\%$ reduction in fuel consumption.

These results highlight the efficiency and sustainability

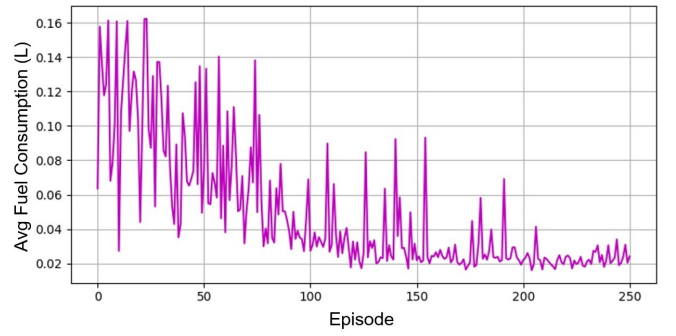


Fig. 5. Average fuel consumption per vehicle (over 250 episodes).

TABLE III
PERFORMANCE COMPARISON OF THE ST-DQN-TA AGENT AGAINST FIXED-TIME TRAFFIC SIGNAL CONTROLLERS AT A STANDARD FOUR-WAY INTERSECTION

Method	Avg Queue	Avg Waiting (s)	Fuel (L)
ST-DQN-TA	4.62	17.31	0.0201
Baseline 30s	21.57	50.41	0.0548
Baseline 40s	31.53	65.64	0.0664
Baseline 50s	39.56	82.94	0.0935
Baseline 60s	47.92	100.24	0.1025

benefits of adaptive, learning-based control over static timing strategies.

V. CONCLUSION

We introduced ST-DQN-TA, a Dual-Stream Spatio-Temporal Deep Q-Network enhanced with Temporal Attention, designed for adaptive signal control at isolated four-way intersections. By capturing both spatial lane-level conditions and temporal traffic evolution, and combining these with multi-head self-attention and a dueling DQN architecture, the model delivers stable and high-quality decisions for phase selection and green timing. Experiments in SUMO show consistent reductions in vehicle waiting time, peak queue length, and fuel consumption compared with fixed-time baselines under diverse traffic scenarios.

Future extensions of ST-DQN-TA aim to scale beyond a single junction, addressing the increased complexity of multi-intersection networks. These extensions will also incorporate priority vehicles (e.g., ambulances, police cars, and fire trucks), pedestrian flows, and crosswalk dynamics, alongside broader benchmarking against cutting-edge DRL controllers. Results from these multi-intersection studies will be presented in future publications to validate the scalability, robustness, and real-world applicability of the proposed framework.

ACKNOWLEDGMENT

This research was supported by the UTAR Research Fund (UTARRF) of Universiti Tunku Abdul Rahman (UTAR). The authors sincerely thank UTAR for the financial support and continuous encouragement that made this work possible.

REFERENCES

- [1] S. Sirphy and S. Thanga Revathi, "Adaptive traffic control system using YOLO," in *Proceedings of the 2023 International Conference on Computer Communication and Informatics (ICCCI)*, pp. 1–5, 2023.
- [2] R. Shyam Shankaran and L. Rajendran, "Real-time adaptive traffic control system for smart cities," in *Proceedings of the 2021 International Conference on Computer Communication and Informatics (ICCCI)*, pp. 1–6, 2021.
- [3] S. G. Devika, A. Govind, and D. Lekshmi, "Next-generation traffic control: Adaptive timer and emergency vehicle priority in intelligent traffic management," in *Proceedings of the 2024 International Conference on E-Mobility, Power Control and Smart Systems (ICEMPS)*, pp. 1–6, 2024.
- [4] A. Yumaganov, A. Agafonov, and V. Myasnikov, "Cooperative application of vehicular traffic rerouting method and adaptive traffic signal control method," in *Proceedings of the 9th International Conference on Information Technology and Nanotechnology (ITNT)*, pp. 1–5, 2023.
- [5] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wiessner, "Microscopic traffic simulation using SUMO," in *Proceedings of the 21st International Conference on Intelligent Transportation Systems*, pp. 2575–2582, 2018.
- [6] A. Wegener, M. Piórkowski, M. Raya, H. Hellbrück, S. Fischer, and J.-P. Hubaux, "TraCI: An interface for coupling road traffic and network simulators," in *Proceedings of the 11th Communications and Networking Simulation Symposium*, pp. 155–163, 2008.
- [7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [8] M. Noeen, A. Naik, L. Goodman, J. Crebo, T. Abrar, Z. Abad, A. Bazzan, and B. Far, "Reinforcement learning in urban network traffic signal control: A systematic literature review," *Expert Systems with Applications*, vol. 199, art. 116830, 2022.
- [9] L. Zhang, S. Xie, and J. Deng, "Leveraging queue length and attention mechanisms for enhanced traffic signal control optimization," in *Proceedings of the Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pp. 141–156, 2023.
- [10] H. Zhang, Z. Fang, Y. Chen, H. Dai, Q. Jiang, and X. Zeng, "Traffic signal optimization control method based on attention mechanism updated weights double deep Q network," *Complex and Intelligent Systems*, vol. 11, no. 5, p. 217, 2025.
- [11] M. Li, X. Pan, C. Liu, and Z. Li, "Federated deep reinforcement learning-based urban traffic signal optimal control," *Scientific Reports*, vol. 15, no. 1, p. 11724, 2025.
- [12] G. Yang, X. Wen, and F. Chen, "Multi-agent deep reinforcement learning with graph attention network for traffic signal control in multiple-intersection urban areas," *Transportation Research Record*, vol. 2679, no. 4, pp. 880–898, 2025.
- [13] C. Cai and M. Wei, "Adaptive urban traffic signal control based on enhanced deep reinforcement learning," *Scientific Reports*, vol. 14, no. 1, p. 14116, 2024.